

Process Control & Instrumentation

□ To impart the basic concepts of process control and instrumentation in process

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5072 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5072.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Chemical Engineering
Course code	5072
Course title	Process Control & Instrumentation
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 (L: 3 T: 1 P: 0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To impart the basic concepts of process control and instrumentation in process

Course outcomes

CO	Official outcome	Study level
CO1	12 Understanding industries. Illustrate the measuring instruments used in	See official syllabus
CO2	19 Understanding process plants. Summarise process control system and final	See official syllabus
CO3	19 Understanding control element.	See official syllabus
CO4	Illustrate analytical instruments. 10 Understanding	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2
CO4	CO4 2 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Outline the elements of process control in industries	4	As specified
Module 2	Illustrate the measuring instruments used in process plants	5	As specified
Module 3	Summarize process control system and final control element	4	As specified
Module 4	Illustrate analytical instruments	3	As specified

MODULE 1

Outline the elements of process control in industries

This module is presented from the official syllabus scope for Process Control & Instrumentation. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Outline the elements of process control in industries
- Official duration: As specified
- Official topic entries captured: 4

- The full source excerpt is preserved below for coverage checking.

2 Understanding chemical plants.

2 Understanding chemical plants. is an official topic in Module 1 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding chemical plants. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding chemical plants. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding chemical plants. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-2 Understanding chemical plants.

പ്രക്രിയയുടെ നിർവ്വചനം/അർത്ഥം, ഘട്ടങ്ങൾ, പ്രയോഗം, technical terms English-ൽ പരിഭാഷിക്കുക.

Outline the control system in a chemical plant.

Outline the control system in a chemical plant. is an official topic in Module 1 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the control system in a chemical plant. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the control system in a chemical plant. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the control system in a chemical plant. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഔദ്യോഗിക നിയന്ത്രണ സംവിധാനം. ഔദ്യോഗിക നിയന്ത്രണ സംവിധാനം, steps, application ഔദ്യോഗിക നിയന്ത്രണ സംവിധാനം. Technical terms English-ഔദ്യോഗിക നിയന്ത്രണ സംവിധാനം.

Outline the functions of measuring instruments. 3 Understanding Comprehend the characteristics of measuring

Outline the functions of measuring instruments. 3 Understanding Comprehend the characteristics of measuring is an official topic in Module 1 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the functions of measuring instruments. 3 Understanding Comprehend the characteristics of measuring using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the functions of measuring instruments. 3 understanding comprehend the characteristics of measuring should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the functions of measuring instruments. 3 Understanding Comprehend the characteristics of measuring means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഓരോന്നും Outline the functions of measuring instruments.

3 Understanding Comprehend the characteristics of measuring instruments

പദാർത്ഥം definition/meaning പദാർത്ഥം പദാർത്ഥം. പദാർത്ഥം പദാർത്ഥം പദാർത്ഥം, steps, application പദാർത്ഥം പദാർത്ഥം പദാർത്ഥം. Technical terms English-ഓരോന്നും പദാർത്ഥം പദാർത്ഥം.

4 Understanding instruments. Summarize the importance of process control in chemical plants - Requirements of instrumentation and process control - process variables - Advantages of instrumentation and process control. Control system- block diagram - controlled variable, measuring element, measured value, controller, set value, error / deviation, desired value, final control element - Outline a simple level control system - Outline a simple temperature control system. Functions of measuring instruments, signaling, transmitting, registering, indicating and recording - Elements of measuring instruments, primary, secondary, manipulating and functioning element. Characteristics of measuring instruments - Static characteristics, accuracy, reproducibility, sensitivity, static error, drift, dead zone and repeatability - Dynamic characteristics, speed of response, fidelity, lag, dynamic error.

4 Understanding instruments. Summarize the importance of process control in chemical plants - Requirements of instrumentation and process control - process variables - Advantages of instrumentation and process control. Control system- block diagram - controlled variable, measuring element, measured value, controller, set value, error / deviation, desired value, final control element - Outline a simple level control system - Outline a simple temperature control system. Functions of measuring instruments, signaling, transmitting, registering, indicating and recording - Elements of measuring instruments, primary, secondary, manipulating and functioning element. Characteristics of measuring instruments - Static characteristics, accuracy, reproducibility, sensitivity, static error, drift, dead zone and repeatability - Dynamic characteristics, speed of response, fidelity, lag, dynamic error. is an official topic in Module 1 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding instruments. Summarize the importance of process control in chemical plants - Requirements of instrumentation and process control - process variables - Advantages of instrumentation and process control. Control system- block diagram - controlled variable, measuring element, measured value, controller, set value, error / deviation, desired value, final control element - Outline a simple level control system - Outline a simple temperature control

system. Functions of measuring instruments, signaling, transmitting, registering, indicating and recording – Elements of measuring instruments, primary, secondary, manipulating and functioning element. Characteristics of measuring instruments - Static characteristics, accuracy, reproducibility, sensitivity, static error, drift, dead zone and repeatability - Dynamic characteristics, speed of response, fidelity, lag, dynamic error. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding instruments. summarize the importance of process control in chemical plants - requirements of instrumentation and process control - process variables - advantages of instrumentation and process control. control system- block diagram - controlled variable, measuring element, measured value, controller, set value, error / deviation, desired value, final control element - outline a simple level control system – outline a simple temperature control system. functions of measuring instruments, signaling, transmitting, registering, indicating and recording – elements of measuring instruments, primary, secondary, manipulating and functioning element. characteristics of measuring instruments - static characteristics, accuracy, reproducibility, sensitivity, static error, drift, dead zone and repeatability - dynamic characteristics, speed of response, fidelity, lag, dynamic error. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding instruments. Summarize the importance of process control in chemical plants - Requirements of instrumentation and process control - process variables - Advantages of instrumentation and process control. Control system- block diagram - controlled variable, measuring element, measured value, controller, set value, error / deviation, desired value, final control element - Outline a simple level control system – Outline a simple temperature control system. Functions of measuring instruments, signaling, transmitting, registering, indicating and recording – Elements of measuring instruments, primary, secondary, manipulating and functioning element. Characteristics of measuring instruments - Static characteristics, accuracy, reproducibility, sensitivity, static error, drift, dead zone and repeatability - Dynamic characteristics, speed of response, fidelity, lag, dynamic error. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-4 Understanding instruments. Summarize the importance of process control in chemical plants - Requirements of instrumentation and process control - process variables - Advantages of instrumentation and process control. Control system- block diagram - controlled variable, measuring element, measured value, controller, set value, error / deviation, desired value, final control element - Outline a simple level control system - Outline a simple temperature control system. Functions of measuring instruments, signaling, transmitting, registering, indicating and recording - Elements of measuring instruments, primary, secondary, manipulating and functioning element. Characteristics of measuring instruments - Static characteristics, accuracy, reproducibility, sensitivity, static error, drift, dead zone and repeatability - Dynamic characteristics, speed of response, fidelity, lag, dynamic error. പരീക്ഷയിൽ ഉപയോഗിക്കേണ്ട പദങ്ങളുടെ definition/meaning പരിശോധിക്കുക. പരീക്ഷയിൽ ഉപയോഗിക്കേണ്ട പദങ്ങൾ, steps, application പരിശോധിക്കുക. Technical terms English-ൽ പരിശോധിക്കുക.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Outline the elements of process control in industries

M1.01	Summarize the importance of process control in chemical plants.	2
Understanding		
M1.02	Outline the control system in a chemical plant.	3
Understanding		
M1.03	Outline the functions of measuring instruments.	3
Understanding		
M1.04	Comprehend the characteristics of measuring instruments.	4
Understanding		

Contents:

Summarize the importance of process control in chemical plants - Requirements of instrumentation and process control - process variables - Advantages of instrumentation and process control.

Control system- block diagram - controlled variable, measuring element, measured value,

controller, set value, error / deviation, desired value, final control element - Outline a

simple level control system – Outline a simple temperature control system.

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Illustrate the measuring instruments used in process plants

This module is presented from the official syllabus scope for Process Control & Instrumentation. Use the topic cards for learning and the source transcript for exact

coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Illustrate the measuring instruments used in process plants
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

4 Understanding plants. Summarize pressure measurement in process

4 Understanding plants. Summarize pressure measurement in process is an official topic in Module 2 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding plants. Summarize pressure measurement in process using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding plants. summarize pressure measurement in process should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding plants. Summarize pressure measurement in process means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2-4 Understanding plants. Summarize pressure measurement in process ധർമ്മം definition/meaning ധർമ്മം. ധർമ്മം ധർമ്മം ധർമ്മം, steps, application ധർമ്മം ധർമ്മം ധർമ്മം. Technical terms English-ധർമ്മം ധർമ്മം.

4 Understanding plants.

4 Understanding plants. is an official topic in Module 2 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding plants. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding plants. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding plants. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-4 Understanding plants. പഠിക്കേണ്ടതെല്ലാം തന്നെ definition/meaning പഠിക്കേണ്ടതാണ്. പഠിക്കേണ്ടതെല്ലാം പഠിക്കേണ്ടതാണ്, steps, application പഠിക്കേണ്ടതെല്ലാം പഠിക്കേണ്ടതാണ്. Technical terms English-ൽ പഠിക്കേണ്ടതെല്ലാം പഠിക്കേണ്ടതാണ്.

Summarize flow measurement in process plants.

Summarize flow measurement in process plants. is an official topic in Module 2 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize flow measurement in process plants. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize flow measurement in process plants. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize flow measurement in process plants. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-
Summarize flow measurement in process plants. definition/meaning . steps, application . Technical terms English-

Summarize level measurement in process plants. 2 Understanding Summarize fluid property measurements in

Summarize level measurement in process plants. 2 Understanding Summarize fluid property measurements in is an official topic in Module 2 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize level measurement in process plants. 2 Understanding Summarize fluid property measurements in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize level measurement in process plants. 2 understanding summarize fluid property measurements in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize level measurement in process plants. 2 Understanding Summarize fluid property measurements in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-
Summarize level measurement in process plants. 2 Understanding Summarize fluid property measurements in
 definition/meaning . . , steps, application . Technical terms English-
 .

4 Understanding process plants. Temperature measuring instruments - Principle and working of - Thermometers, filled system and vapor pressure thermometers - Thermocouples, types, materials - Resistance thermometers, thermistor - Pyrometer, radiation and optical. Pressure measuring instruments - Principle and working of - Bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. Flow measuring devices: Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement flow meters - Differential pressure transducer used in venturi and orifice meters. Level measuring devices: Float level indicator, Magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. PH measurement, pH scale, reference and measuring electrode - Conductivity measurement, electrical conductivity meter - Density measurement, online density meter, hydrometer. Humidity measurement, dew point method, wet bulb method, hygrometer.

4 Understanding process plants. Temperature measuring instruments - Principle and working of - Thermometers, filled system and vapor pressure thermometers - Thermocouples, types, materials - Resistance thermometers, thermistor - Pyrometer, radiation and optical. Pressure measuring instruments - Principle and working of - Bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. Flow measuring devices: Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement flow meters - Differential pressure transducer used in venturi and orifice meters. Level measuring devices: Float level indicator, Magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. PH measurement, pH scale, reference and measuring electrode - Conductivity measurement, electrical conductivity meter - Density measurement, online density meter, hydrometer. Humidity measurement, dew point method, wet bulb method, hygrometer. is an official topic in Module 2 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding process plants. Temperature measuring instruments - Principle and working of – Thermometers, filled system and vapor pressure thermometers – Thermocouples, types, materials - Resistance thermometers, thermistor – Pyrometer, radiation and optical. Pressure measuring instruments - Principle and working of - Bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. Flow measuring devices: Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement flow meters – Differential pressure transducer used in venturi and orifice meters. Level measuring devices: Float level indicator, Magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. PH measurement, pH scale, reference and measuring electrode - Conductivity measurement, electrical conductivity meter - Density measurement, online density meter, hydrometer. Humidity measurement, dew point method, wet bulb method, hygrometer. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding process plants. temperature measuring instruments - principle and working of – thermometers, filled system and vapor pressure thermometers – thermocouples, types, materials - resistance thermometers, thermistor – pyrometer, radiation and optical. pressure measuring instruments - principle and working of - bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. flow measuring devices: coriolis mass flow meter, ultra sonic flow meter, and positive displacement flow meters – differential pressure transducer used in venturi and orifice meters. level measuring devices: float level indicator, magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. ph measurement, ph scale, reference and measuring electrode - conductivity measurement, electrical conductivity meter - density measurement, online density meter, hydrometer. humidity measurement, dew point method, wet bulb method, hygrometer. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding process plants. Temperature measuring instruments - Principle and working of – Thermometers, filled system and vapor pressure thermometers – Thermocouples, types, materials - Resistance thermometers, thermistor – Pyrometer, radiation and optical. Pressure measuring instruments - Principle and working of - Bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. Flow measuring devices: Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement flow meters – Differential pressure

Aspect	What to prepare
	transducer used in venturi and orifice meters. Level measuring devices: Float level indicator, Magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. PH measurement, pH scale, reference and measuring electrode - Conductivity measurement, electrical conductivity meter - Density measurement, online density meter, hydrometer. Humidity measurement, dew point method, wet bulb method, hygrometer. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-4 Understanding process plants. Temperature measuring instruments - Principle and working of - Thermometers, filled system and vapor pressure thermometers -Thermocouples, types, materials - Resistance thermometers, thermistor - Pyrometer, radiation and optical. Pressure measuring instruments - Principle and working of - Bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. Flow measuring devices: Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement flow meters - Differential pressure transducer used in venturi and orifice meters. Level measuring devices: Float level indicator, Magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. PH measurement, pH scale, reference and measuring electrode - Conductivity measurement, electrical conductivity meter - Density measurement, online density meter, hydrometer. Humidity measurement, dew point method, wet bulb method, hygrometer. താപനില അളക്കുന്ന ഉപകരണങ്ങൾ definition/meaning താപനില അളക്കുന്ന ഉപകരണങ്ങൾ. താപനില അളക്കുന്ന ഉപകരണങ്ങൾ, steps, application താപനില അളക്കുന്ന ഉപകരണങ്ങൾ. Technical terms English- താപനില അളക്കുന്ന ഉപകരണങ്ങൾ.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Illustrate the measuring instruments used in process plants

M2.01	Summarize temperature measurement in process	4
Understanding	plants.	
M2.02	Summarize pressure measurement in process	4
Understanding	plants.	
M2.03	Summarize flow measurement in process plants.	4
Understanding		
M2.04	Summarize level measurement in process plants.	2
Understanding		
M2.05	Summarize fluid property measurements in	4
Understanding	process plants.	
	Series Test – I	1

Contents:

Temperature measuring instruments - Principle and working of – Thermometers, filled system and vapor pressure thermometers – Thermocouples, types, materials - Resistance thermometers, thermistor – Pyrometer, radiation and optical.

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Summarize process control system and final control element

This module is presented from the official syllabus scope for Process Control & Instrumentation. Use the topic cards for learning and the source transcript for exact

coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize process control system and final control element
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

6 Understanding control.

6 Understanding control. is an official topic in Module 3 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding control. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding control. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding control. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-6 Understanding control. ഏകീകൃതമായ നിയന്ത്രണ
definition/meaning നിയന്ത്രണത്തിന്റെ അർത്ഥം. നിയന്ത്രണത്തിന്റെ ഘട്ടങ്ങൾ, steps, application
നിയന്ത്രണത്തിന്റെ പ്രയോഗം. Technical terms English-നിയന്ത്രണത്തിന്റെ പദങ്ങൾ.

Summarize computerized control system.

Summarize computerized control system. is an official topic in Module 3 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize computerized control system. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize computerized control system. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize computerized control system. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഉം Summarize computerized control system.

കമ്പ്യൂട്ടറൈസ്ഡ് കൺട്രോൾ സിസ്റ്റം definition/meaning ഉൾപ്പെടെ. കമ്പ്യൂട്ടറൈസ്ഡ് കൺട്രോൾ സിസ്റ്റം, steps, application ഉൾപ്പെടെ. Technical terms English-ൽ ഉൾപ്പെടെ ഉൾപ്പെടെ.

Summarize final control elements.

Summarize final control elements. is an official topic in Module 3 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize final control elements. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize final control elements. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize final control elements. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-എന്നത് Summarize final control elements.

എന്നതിന്റെ definition/meaning എഴുതുക. എന്നതിന്റെ steps, application എന്നിവ എഴുതുക. Technical terms English-ൽ എഴുതുക.

Outline instrumented safety system. 3 Understanding Illustrate On-Off (Two position control), feedback and feed forward control systems - Outline servo and regulatory control - Illustrate Proportional (P), integral (I), derivative (D) and PID control - Illustrate special type of controllers, cascade control, ratio of control and split range control - Sketch behaviour of different modes of control. Outline analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter - Illustrate valve safe positions - Fail open - Fail close - Fail steady - Variable frequency drive [VFD] Safety Instrumentation System - Emergency shutdown system - Safety interlocks.

Outline instrumented safety system. 3 Understanding Illustrate On-Off (Two position control), feedback and feed forward control systems - Outline servo and regulatory control - Illustrate Proportional (P), integral (I), derivative (D) and PID control - Illustrate special type of controllers, cascade control, ratio of control and split range control - Sketch behaviour of different modes of control. Outline analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter - Illustrate valve safe positions - Fail open - Fail close - Fail steady - Variable frequency drive [VFD] Safety Instrumentation System - Emergency shutdown system - Safety interlocks. is an official topic in Module 3 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline instrumented safety system. 3 Understanding Illustrate On-Off (Two position control), feedback and feed forward control systems – Outline servo and regulatory control - Illustrate Proportional (P), integral (I), derivative (D) and PID control - Illustrate special type of controllers, cascade control, ratio of control and split range control - Sketch behaviour of different modes of control. Outline analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter - Illustrate valve safe positions - Fail open - Fail close - Fail steady - Variable frequency drive [VFD] Safety Instrumentation System – Emergency shutdown system – Safety interlocks. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline instrumented safety system. 3 understanding illustrate on-off (two position control), feedback and feed forward control systems – outline servo and regulatory control - illustrate proportional (p), integral (i), derivative (d) and pid control - illustrate special type of controllers, cascade control, ratio of control and split range control - sketch behaviour of different modes of control. outline analog controllers, electronic controllers and pneumatic controllers. outline computerized control in process plants - programmable logic controllers - supervisory control and data acquisition (scada) - rtu (remote terminal unit) - distributed control system (dcs). final control elements - mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, i to p converter - illustrate valve safe positions - fail open - fail close - fail steady - variable frequency drive [vfd] safety instrumentation system – emergency shutdown system – safety interlocks. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline instrumented safety system. 3 Understanding Illustrate On-Off (Two position control), feedback and feed forward control systems – Outline servo and regulatory control - Illustrate Proportional (P), integral (I), derivative (D) and PID control - Illustrate special type of controllers, cascade control, ratio of control and split range control - Sketch behaviour of different modes of control. Outline analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system

Aspect	What to prepare
	(DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter - Illustrate valve safe positions - Fail open - Fail close - Fail steady - Variable frequency drive [VFD] Safety Instrumentation System - Emergency shutdown system - Safety interlocks. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓൗട്ട്ലൈൻ ഇൻസ്ട്രുമെന്റേഡ് സേഫ്റ്റി സിസ്റ്റം. 3 Understanding Illustrate On-Off (Two position control), feedback and feed forward control systems - Outline servo and regulatory control - Illustrate Proportional (P), integral (I), derivative (D) and PID control - Illustrate special type of controllers, cascade control, ratio of control and split range control - Sketch behaviour of different modes of control. Outline analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter - Illustrate valve safe positions - Fail open - Fail close - Fail steady - Variable frequency drive [VFD] Safety Instrumentation System - Emergency shutdown system - Safety interlocks. ഓൗട്ട്ലൈൻ definition/meaning ഓൗട്ട്ലൈൻ. ഓൗട്ട്ലൈൻ ഓൗട്ട്ലൈൻ ഓൗട്ട്ലൈൻ, steps, application ഓൗട്ട്ലൈൻ ഓൗട്ട്ലൈൻ ഓൗട്ട്ലൈൻ. Technical terms English-ഓൗട്ട്ലൈൻ ഓൗട്ട്ലൈൻ.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Summarize process control system and final control element

	Summarize the principles of automatic process control.	
M3.01		6
Understanding		
	Summarize computerized control system.	
M3.02		5
Understanding		
	Summarize final control elements.	
M3.03		5
Understanding		
	Outline instrumented safety system.	
M3.04		3
Understanding		

Contents:

Illustrate On-Off (Two position control), feedback and feed forward control systems –

Outline servo and regulatory control - Illustrate Proportional (P), integral (I), derivative (D)

and PID control - Illustrate special type of controllers, cascade control, ratio of control and

split range control - Sketch behaviour of different modes of control. Outline analog

controllers, electronic controllers and pneumatic controllers.

Outline computerized Control in process plants - Programmable logic controllers -

Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) -

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Illustrate analytical instruments

This module is presented from the official syllabus scope for Process Control & Instrumentation. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Illustrate analytical instruments
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Summarize chromatographic methods of analysis.

Summarize chromatographic methods of analysis. is an official topic in Module 4 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize chromatographic methods of analysis. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize chromatographic methods of analysis. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize chromatographic methods of analysis. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-
Summarize chromatographic methods of analysis.
definition/meaning .
steps, application . Technical terms English-
.

Outline spectroscopic methods of analysis.

Outline spectroscopic methods of analysis. is an official topic in Module 4 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline spectroscopic methods of analysis. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline spectroscopic methods of analysis. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline spectroscopic methods of analysis. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഔട്ലൈൻ സ്പെക്ട്രോസ്കോപ്പിക് വിശ്ലേഷണ രീതികൾ.

ഔട്ട്ലൈൻ സ്പെക്ട്രോസ്കോപ്പിക് വിശ്ലേഷണ രീതികൾ definition/meaning ഔട്ട്ലൈൻ സ്പെക്ട്രോസ്കോപ്പിക് വിശ്ലേഷണ രീതികൾ. ഔട്ട്ലൈൻ സ്പെക്ട്രോസ്കോപ്പിക് വിശ്ലേഷണ രീതികൾ, steps, application ഔട്ട്ലൈൻ സ്പെക്ട്രോസ്കോപ്പിക് വിശ്ലേഷണ രീതികൾ. Technical terms English-ഔട്ട്ലൈൻ സ്പെക്ട്രോസ്കോപ്പിക് വിശ്ലേഷണ രീതികൾ.

Summarize structural analysis 3 Understanding Chromatographic analysis - Working and applications of - Liquid solid chromatography - Gas solid chromatography - Liquid liquid chromatography - High performance liquid chromatography. Spectroscopy - Outline the classification of spectrometers - Illustrate the working of uv- visible absorption spectrometer - Atomic emission spectrometer - Mass spectrometer. Structure analysis - Outline SEM (scanning electron microscope), TEM (Transmission electron microscope), XRD (X-ray diffraction). Group activity (Prepare a literature review report on the usage of SEM, TEM and XRD in analyzing nanoparticles) Text / Reference T/R Book Title/Author T1 Outlines of chemical instrumentation & Process control; Dr. A. Sooryanarayana - R1 Mechanical& Industrial Measurements; R.K Jain R2 Industrial Instrumentation ; Donald P. Eckman - R3 Engineering Chemistry; Jain & Jain R4 Industrial instrumentation ; K Krishnaswami - R5 Instrumentation and Process Control; D.C. Sikdar, Online Resources S.No Website Link 1 <https://www.nptel.ac.in/courses/103105064>

Summarize structural analysis 3 Understanding Chromatographic analysis - Working and applications of - Liquid solid chromatography - Gas solid chromatography - Liquid liquid chromatography - High performance liquid chromatography. Spectroscopy - Outline the classification of spectrometers - Illustrate the working of uv- visible absorption spectrometer - Atomic emission spectrometer - Mass spectrometer. Structure analysis - Outline SEM (scanning electron microscope), TEM (Transmission electron microscope), XRD (X-ray diffraction). Group activity (Prepare a literature review report on the usage of SEM, TEM and XRD in analyzing nanoparticles) Text / Reference T/R Book Title/Author T1 Outlines of chemical instrumentation & Process control; Dr. A. Sooryanarayana - R1 Mechanical& Industrial Measurements; R.K Jain R2 Industrial Instrumentation ; Donald P. Eckman - R3 Engineering Chemistry; Jain & Jain R4 Industrial instrumentation ; K Krishnaswami - R5 Instrumentation and Process Control; D.C. Sikdar, Online Resources S.No Website Link 1 <https://www.nptel.ac.in/courses/103105064> is an official topic in Module 4 of Process Control & Instrumentation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize structural analysis 3 Understanding Chromatographic analysis – Working and applications of - Liquid solid chromatography - Gas solid chromatography - Liquid liquid chromatography - High performance liquid chromatography. Spectroscopy – Outline the classification of spectrometers – Illustrate the working of uv- visible absorption spectrometer - Atomic emission spectrometer - Mass spectrometer. Structure analysis – Outline SEM (scanning electron microscope), TEM (Transmission electron microscope), XRD (X-ray diffraction). Group activity (Prepare a literature review report on the usage of SEM, TEM and XRD in analyzing nanoparticles) Text / Reference T/R Book Title/Author T1 Outlines of chemical instrumentation & Process control; Dr. A. Sooryanarayana - R1 Mechanical& Industrial Measurements; R.K Jain R2 Industrial Instrumentation ; Donald P. Eckman - R3 Engineering Chemistry; Jain & Jain R4 Industrial instrumentation ; K Krishnaswami - R5 Instrumentation and Process Control; D.C. Sikdar, Online Resources S.No Website Link 1 <https://www.nptel.ac.in/courses/103105064> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize structural analysis 3 understanding chromatographic analysis – working and applications of - liquid solid chromatography - gas solid chromatography - liquid liquid chromatography - high performance liquid chromatography. spectroscopy – outline the classification of spectrometers – illustrate the working of uv- visible absorption spectrometer - atomic emission spectrometer - mass spectrometer. structure analysis – outline sem (scanning electron microscope), tem (transmission electron microscope), xrd (x-ray diffraction). group activity (prepare a literature review report on the usage of sem, tem and xrd in analyzing nanoparticles) text / reference t/r book title/author t1 outlines of chemical instrumentation & process control; dr. a. sooryanarayana - r1 mechanical& industrial measurements; r.k jain r2 industrial instrumentation ; donald p. eckman - r3 engineering chemistry; jain & jain r4 industrial instrumentation ; k krishnaswami - r5 instrumentation and process control; d.c. sikdar, online resources s.no website link 1 <https://www.nptel.ac.in/courses/103105064> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Summarize structural analysis 3 Understanding Chromatographic analysis – Working and applications of - Liquid solid chromatography - Gas solid chromatography - Liquid liquid chromatography - High performance liquid chromatography. Spectroscopy – Outline the classification of spectrometers – Illustrate the working of uv- visible absorption spectrometer - Atomic emission spectrometer - Mass spectrometer. Structure analysis – Outline SEM (scanning electron microscope), TEM (Transmission electron microscope), XRD (X-ray diffraction). Group activity (Prepare a literature review report on the usage of SEM, TEM and XRD in analyzing nanoparticles) Text / Reference T/R Book Title/Author T1 Outlines of chemical instrumentation & Process control; Dr. A. Sooryanarayana - R1 Mechanical& Industrial Measurements; R.K Jain R2 Industrial Instrumentation ; Donald P. Eckman - R3 Engineering Chemistry; Jain & Jain R4 Industrial instrumentation ; K Krishnaswami - R5 Instrumentation and Process Control; D.C. Sikdar, Online Resources S.No Website Link 1 https://www.nptel.ac.in/courses/103105064 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Summarize structural analysis 3 Understanding Chromatographic analysis – Working and applications of - Liquid solid chromatography - Gas solid chromatography - Liquid liquid chromatography - High performance liquid chromatography. Spectroscopy – Outline the classification of spectrometers – Illustrate the working of uv- visible absorption spectrometer - Atomic emission spectrometer - Mass spectrometer. Structure analysis – Outline SEM (scanning electron microscope), TEM (Transmission electron microscope), XRD (X-ray diffraction). Group activity (Prepare a literature review report on the usage of SEM, TEM and XRD in analyzing nanoparticles) Text / Reference T/R Book Title/Author T1 Outlines of chemical instrumentation & Process control; Dr. A. Sooryanarayana - R1 Mechanical& Industrial Measurements; R.K Jain R2 Industrial Instrumentation ; Donald P. Eckman - R3 Engineering Chemistry; Jain & Jain R4 Industrial instrumentation ; K Krishnaswami - R5 Instrumentation and Process Control; D.C. Sikdar, Online Resources S.No Website Link 1 <https://www.nptel.ac.in/courses/103105064> definition/meaning . , steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Illustrate analytical instruments

M4.01 Summarize chromatographic methods of analysis. 3
Understanding

M4.02 Outline spectroscopic methods of analysis. 3
Understanding

M4.03 Summarize structural analysis 3
Understanding

Series Test – II 1

Contents:

Chromatographic analysis – Working and applications of - Liquid solid chromatography -

Gas solid chromatography - Liquid liquid chromatography - High performance liquid chromatography.

Spectroscopy – Outline the classification of spectrometers – Illustrate the working of uv-visible absorption spectrometer - Atomic emission spectrometer - Mass spectrometer.

Structure analysis – Outline SEM (scanning electron microscope), TEM (Transmission electron microscope), XRD (X-ray diffraction).

Group activity (Prepare a literature review report on the usage of SEM, TEM and XRD in

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 2 Understanding chemical plants.. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding chemical plants..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Outline the control system in a chemical plant.. (3 marks)

Answer: Begin with the standard definition or identity of *Outline the control system in a chemical plant..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Outline the functions of measuring instruments. 3 Understanding Comprehend the characteristics of measuring. (3 marks)

Answer: Begin with the standard definition or identity of *Outline the functions of measuring instruments. 3 Understanding Comprehend the characteristics of measuring.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Understanding instruments. Summarize the importance of process control in chemical plants - Requirements of instrumentation and process control - process variables - Advantages of instrumentation and process control. Control system- block diagram - controlled variable, measuring element, measured value, controller, set value, error / deviation, desired value, final control element - Outline a simple level control system - Outline a simple temperature control system. Functions of measuring instruments, signaling, transmitting, registering, indicating and recording - Elements of measuring instruments, primary, secondary, manipulating and functioning element. Characteristics of measuring instruments - Static characteristics, accuracy, reproducibility, sensitivity, static error, drift, dead

zone and repeatability - Dynamic characteristics, speed of response, fidelity, lag, dynamic error.. (3 marks)

Answer: Begin with the standard definition or identity of 4 *Understanding instruments*. Summarize the importance of process control in chemical plants - Requirements of instrumentation and process control - process variables - Advantages of instrumentation and process control. Control system- block diagram - controlled variable, measuring element, measured value, controller, set value, error / deviation, desired value, final control element - Outline a simple level control system - Outline a simple temperature control system. Functions of measuring instruments, signaling, transmitting, registering, indicating and recording - Elements of measuring instruments, primary, secondary, manipulating and functioning element. Characteristics of measuring instruments - Static characteristics, accuracy, reproducibility, sensitivity, static error, drift, dead zone and repeatability - Dynamic characteristics, speed of response, fidelity, lag, dynamic error.. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 2

Define or explain 4 Understanding plants. Summarize pressure measurement in process. (3 marks)

Answer: Begin with the standard definition or identity of 4 *Understanding plants*. Summarize pressure measurement in process. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Understanding plants.. (3 marks)

Answer: Begin with the standard definition or identity of 4 *Understanding plants*.. State the governing principle, list the key components or stages, and close with one relevant

application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Summarize flow measurement in process plants.. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize flow measurement in process plants..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Summarize level measurement in process plants. 2 Understanding Summarize fluid property measurements in. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize level measurement in process plants. 2 Understanding Summarize fluid property measurements in.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain 6 Understanding control.. (3 marks)

Answer: Begin with the standard definition or identity of *6 Understanding control..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Summarize computerized control system.. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize computerized control system..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Summarize final control elements.. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize final control elements..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Outline instrumented safety system. 3 Understanding Illustrate On-Off (Two position control), feedback and feed forward control systems - Outline servo and regulatory control - Illustrate Proportional (P), integral (I), derivative (D) and PID control - Illustrate special type of controllers, cascade control, ratio of control and split range control - Sketch behaviour of different modes of control. Outline analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter - Illustrate valve safe positions - Fail open - Fail close - Fail steady - Variable frequency drive [VFD] Safety Instrumentation System - Emergency shutdown system - Safety interlocks.. (3 marks)

Answer: Begin with the standard definition or identity of *Outline instrumented safety system*. 3 *Understanding Illustrate On-Off (Two position control), feedback and feed forward control systems - Outline servo and regulatory control - Illustrate Proportional (P), integral (I), derivative (D) and PID control - Illustrate special type of controllers, cascade control, ratio of control and split range control - Sketch behaviour of different modes of control. Outline analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter - Illustrate valve safe positions - Fail open - Fail close - Fail steady - Variable frequency drive [VFD] Safety Instrumentation System - Emergency shutdown system - Safety interlocks.. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.*

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 4

Define or explain Summarize chromatographic methods of analysis.. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize chromatographic methods of analysis..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Outline spectroscopic methods of analysis.. (3 marks)

Answer: Begin with the standard definition or identity of *Outline spectroscopic methods of analysis..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Summarize structural analysis 3 Understanding Chromatographic analysis - Working and applications of - Liquid solid chromatography - Gas solid chromatography - Liquid liquid chromatography - High performance liquid chromatography. Spectroscopy - Outline the classification of spectrometers - Illustrate the working of uv- visible absorption spectrometer - Atomic emission spectrometer - Mass spectrometer. Structure analysis - Outline SEM (scanning electron microscope), TEM (Transmission electron microscope), XRD (X-ray diffraction). Group activity (Prepare a literature review report on the usage of SEM, TEM and XRD in analyzing nanoparticles) Text / Reference T/R Book Title/Author T1 Outlines of chemical instrumentation & Process control; Dr. A. Sooryanarayana - R1 Mechanical& Industrial Measurements; R.K Jain R2 Industrial Instrumentation ; Donald P. Eckman - R3 Engineering Chemistry; Jain & Jain R4 Industrial instrumentation ; K Krishnaswami - R5 Instrumentation and Process Control; D.C. Sikdar, Online Resources S.No Website Link 1 <https://www.nptel.ac.in/courses/103105064>. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize structural analysis 3 Understanding Chromatographic analysis - Working and applications of - Liquid solid chromatography - Gas solid chromatography - Liquid liquid chromatography - High performance liquid chromatography. Spectroscopy - Outline the classification of spectrometers - Illustrate the working of uv- visible absorption spectrometer - Atomic emission spectrometer - Mass spectrometer. Structure analysis - Outline SEM (scanning electron microscope), TEM (Transmission electron microscope), XRD (X-ray diffraction). Group activity (Prepare a literature review report on the usage of SEM, TEM and XRD in analyzing nanoparticles) Text / Reference T/R Book Title/Author T1 Outlines of chemical instrumentation & Process control; Dr. A. Sooryanarayana - R1 Mechanical& Industrial Measurements; R.K Jain R2 Industrial Instrumentation ; Donald P. Eckman - R3 Engineering Chemistry; Jain & Jain R4 Industrial instrumentation ; K Krishnaswami - R5 Instrumentation and Process Control; D.C. Sikdar, Online Resources S.No Website Link 1 <https://www.nptel.ac.in/courses/103105064>. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.*

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **2 Understanding chemical plants..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **4 Understanding plants. Summarize pressure measurement in process.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **6 Understanding control..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Summarize chromatographic methods of analysis**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
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Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- S.No Website Link <https://www.nptel.ac.in/courses/103105064>

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTTR syllabus PDF for Course 5072. Verify institutional updates against the current official curriculum.